



Post-registration and post-marketing monitoring: Most frequently reported are the bleeding symptoms, mainly in the first month of therapy.

The following undesirable effects are reported, which depending on their frequency, are defined as "very rare":

Blood and lymph system:

- very rare: thrombotic thrombocytopenic purpura (TTP), changes in the count of cells of various blood lines (severe thrombocytopenia, granulocytopenia, agranulocytosis, anemia, aplastic anemia/pancitopenia)

Immune system:

- very rare: anaphylactoid reactions

Mental:

- very rare: confusion, hallucinations.

Nervous system:

- very rare: taste disorder

Vascular system:

- very rare: vasculitis (vascular inflammation), hypotension (low arterial pressure)

Gastrointestinal system:

- rare: colitis (incl. ulcerous or lymphocyte colitis), pancreatitis, stomatitis.

Hepatic:

- Very rare: hepatitis, acute hepatic failure.

Skin and hypodermis:

- Very rare: angioedema, bullous dermatitis (erythema multiforme, Stevens-Johnson syndrome), erythematous rash, urticaria, eczema, lichen planus.

Musculoskeletal:

- very rare: arthritis, muscular and articular pains

Renal:

- Very rare: glomerulonephritis

General:

- Very rare: fever.

Laboratory changes:

- Very rare: elevated values of the indices of hepatic function, elevated serum creatinine.

Overdose

Overdose in treatment with clopidogrel may provoke prolongation of bleeding time, which may lead to hemorrhagic complications. With regards to mechanism of its action, it may be expected positive effect at infusion of thrombocytic mass. There is no specific antidote.

Pharmacodynamic properties

Pharmacotherapeutical group: platelet aggregation inhibitors excl. Heparin, ATC Code: B01AC/04.

Clopidogrel selectively inhibits the binding of adenosine diphosphate (ADP) to its platelet receptor, and the subsequent ADP-mediated activation of the GPIIb/IIIa complex, thereby inhibiting platelet aggregation. Biotransformation of clopidogrel is necessary to produce inhibition of platelet aggregation. Clopidogrel also inhibits platelet aggregation induced by other agonists by blocking the amplification of platelet activation by released ADP. Clopidogrel acts by irreversibly modifying the platelet ADP receptor. Consequently, platelets exposed to clopidogrel are affected for the remainder of their lifespan and recovery of normal platelet function occurs at a rate consistent with platelet turnover. Repeated doses of 75 mg per day produced substantial inhibition of ADP-induced platelet aggregation from the first day; this increased progressively and reached steady state between Day 3 and Day 7. At steady state, the average inhibition level observed with a dose of 75 mg per day was between 40% and 60%. Platelet aggregation and bleeding time gradually returned to baseline values, generally within 5 days after treatment was discontinued.

Pharmacokinetic properties

After repeated doses of 75 mg per day, clopidogrel is rapidly absorbed. However, plasma concentrations of the parent compound are very low and below the quantification limit (0.00025 mg/l) beyond 2 hours. Absorption is at least 50%, based on urinary excretion of clopidogrel metabolites. Clopidogrel is extensively metabolised by the liver and the main metabolite, which is inactive, is the carboxylic acid derivative, which represents about 85% of the circulating compound in plasma. Peak plasma levels of this metabolite (approx. 3 mg/l after repeated 75 mg oral doses) occurred approximately 1 hour after dosing.

Clopidogrel is a prodrug. The active metabolite, a thiol derivative, is formed by oxidation of clopidogrel to 2-oxo-clopidogrel and subsequent hydrolysis. The oxidative step is regulated primarily by Cytochrome P450 isoenzymes 2B6 and 3A4 and to a lesser extent by 1A1, 1A2 and 2C19. The active thiol metabolite, which has been isolated in vitro, binds rapidly and irreversibly to platelet receptors, thus inhibiting platelet aggregation. This metabolite has not been detected in plasma.

The kinetics of the main circulating metabolite were linear (plasma concentrations increased in proportion to dose) in the dose range of 50 to 150 mg of clopidogrel.

Clopidogrel and the main circulating metabolite bind reversibly in vitro to human plasma proteins (98% and 94% respectively). The binding is non-saturable in vitro over a wide concentration range. Following an oral dose of ¹⁴C-labelled clopidogrel in man, approximately 50% was excreted in the urine and approximately 46% in the faeces in the 120-hour interval after dosing. The elimination half-life of the main circulating metabolite was 8 hours after single and repeated administration.

After repeated doses of 75 mg clopidogrel per day, plasma levels of the main circulating metabolite were lower in subjects with

severe renal disease, (creatinine clearance from 5 to 15 ml/min) compared to subjects with moderate renal disease (creatinine clearance from 30 to 60 ml/min) and to levels observed in other studies with healthy subjects. Although inhibition of ADP-induced platelet aggregation was lower (25%) than that observed in healthy subjects, the prolongation of bleeding was similar to that seen in healthy subjects receiving 75 mg of clopidogrel per day. In addition, clinical tolerance was good in all patients.

The pharmacokinetics and pharmacodynamics of clopidogrel were assessed in a single and multiple dose study in both healthy subjects and those with cirrhosis (Child-Pugh Class A or B). Daily dosing for 10 days with clopidogrel 75 mg/day was safe and well tolerated. Clopidogrel C_{max} for both single dose and steady state for cirrhotics was many fold higher than in normal subjects. However, plasma levels of the main circulating metabolite together with the effect of clopidogrel on ADP-induced platelet aggregation and bleeding time were comparable between these groups.

List of excipients

Lactose anhydrous, Cellulose Microcrystalline, Crospovidone (type A), Glycerol dibehenate, Talc.

Composition of the film-coating agent: Polyvinyl alcohol, Talc, Titanium dioxide (E171), Macrogol 3350, Lecithin (E322), Iron oxide red (E171).

Incompatibilities

Not applicable.

Shelf life

36 months

Special precautions for storage

Do not store above 30°C.

Nature and contents of container

Blisters (aluminium/aluminium).

Pack size: Blister packs of 3x10's and 10x10's.

Special precautions for disposal

No special requirements.

Manufacturer

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Date of (Partial) Revision of the Text

November 2021

Actavis

AAAN9551
1022815-04 (10454)